

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Scenario-Based

Course Code:	ITA0512	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Scenario-Based
Weightage:	10% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)</i>		
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Batch	Third Year/ B. Tech IT	Date:	07/08/26

Code of Conduct

I ___Kaviya V / 192421022___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___Kaviya V___

RUBRICS FOR CALCULATION: ASSESSMENT TOOL 2: Scenario-Based

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Problem

Object Boundary Detection

Scenario: *In a medical image, boundaries between tissues need to be clearly identified.*

(a) Understanding of the Scenario

Medical images such as MRI, CT scans, and X-rays contain different tissues, organs, and possible abnormalities. Accurate identification of tissue boundaries is essential because doctors rely on these boundaries to distinguish healthy tissues from diseased regions. Without proper boundary detection, tissues may overlap visually, making diagnosis difficult. Therefore, object boundary detection is a critical preprocessing step in medical image analysis.

Importance:

- Identifies boundaries between different tissues.
- Helps separate healthy and abnormal regions.
- Improves image segmentation accuracy.
- Assists doctors in diagnosis and treatment planning.
- Enables accurate measurement of organs and tumors.

(b) Identification of Key Issues:

Several challenges affect boundary detection in medical images:

1. **Image Noise** - Random noise generated during image acquisition may create false edges.
2. **Low Contrast** - Different tissues often have similar intensity values, making boundaries difficult to distinguish.
3. **Blurred Boundaries** - Patient movement or imaging limitations can produce unclear edges.
4. **Complex Tissue Structures** - Overlapping organs and irregular tissue shapes complicate boundary detection.
5. **Uneven Illumination** - Variations in brightness reduce the visibility of true boundaries.

(c) Application of Concepts:

Selected Method: Canny Edge Detection

Definition: The Canny Edge Detection algorithm is chosen because it provides accurate and continuous boundary detection while minimizing the effects of noise.

Working Steps:

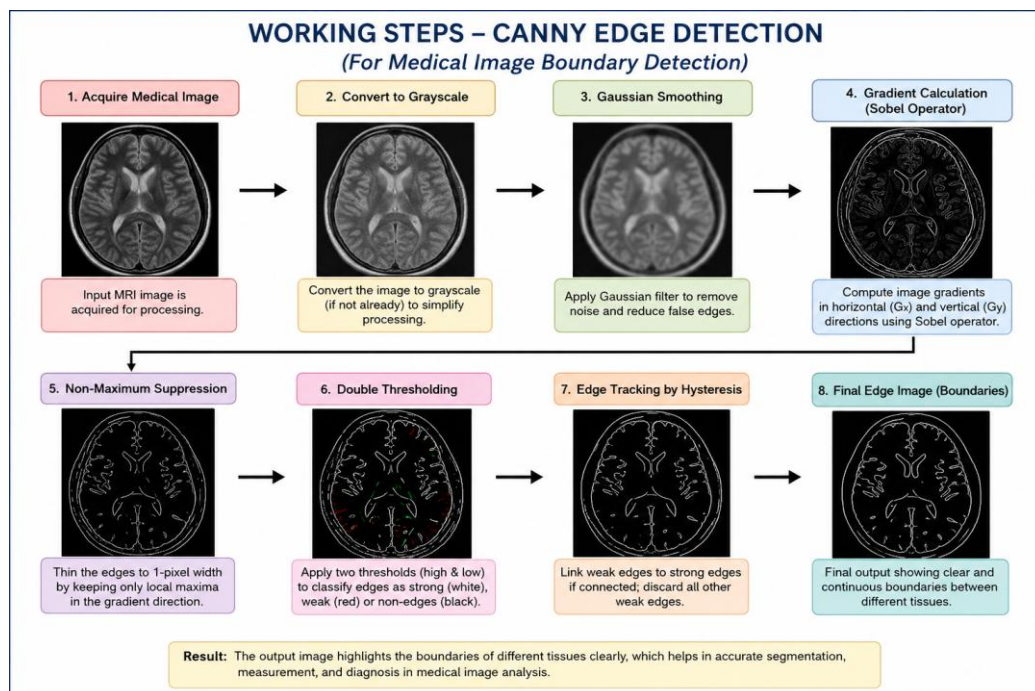
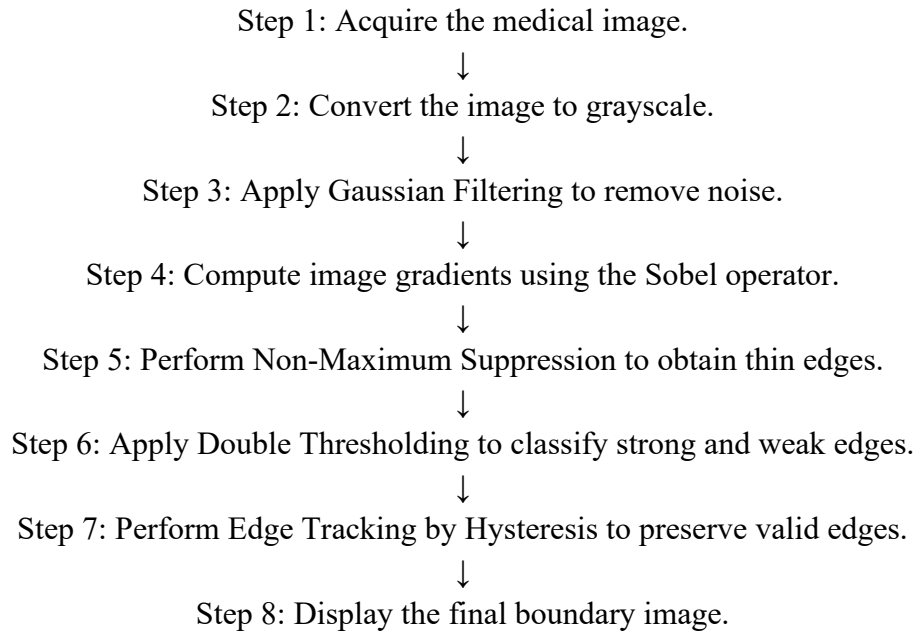


Figure 1: Working steps of Canny Edge Detection.

(d) Decision Making and Justification:

Among different edge detection methods such as Sobel, Prewitt, Roberts, and Canny, the Canny Edge Detection algorithm is the most suitable for medical imaging.

Reasoning:

Method	Limitation
Roberts	Sensitive to noise
Prewitt	Produces thick edges
Sobel	Detects edges but is affected by noise
Canny	Produces thin, continuous, and accurate edges with noise reduction

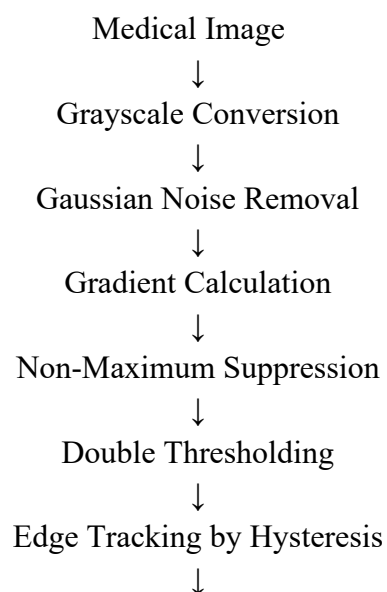
Why Canny?

- Performs noise reduction before edge detection.
- Detects both strong and weak edges.
- Produces thin and continuous boundaries.
- Reduces false edge detection.
- Widely used in MRI, CT scan, and X-ray image analysis.

Decision: Therefore, Canny Edge Detection is the best choice for detecting tissue boundaries in medical images.

(e) Clear and Systematic Explanation:

Flow Diagram:



Detected Tissue Boundaries

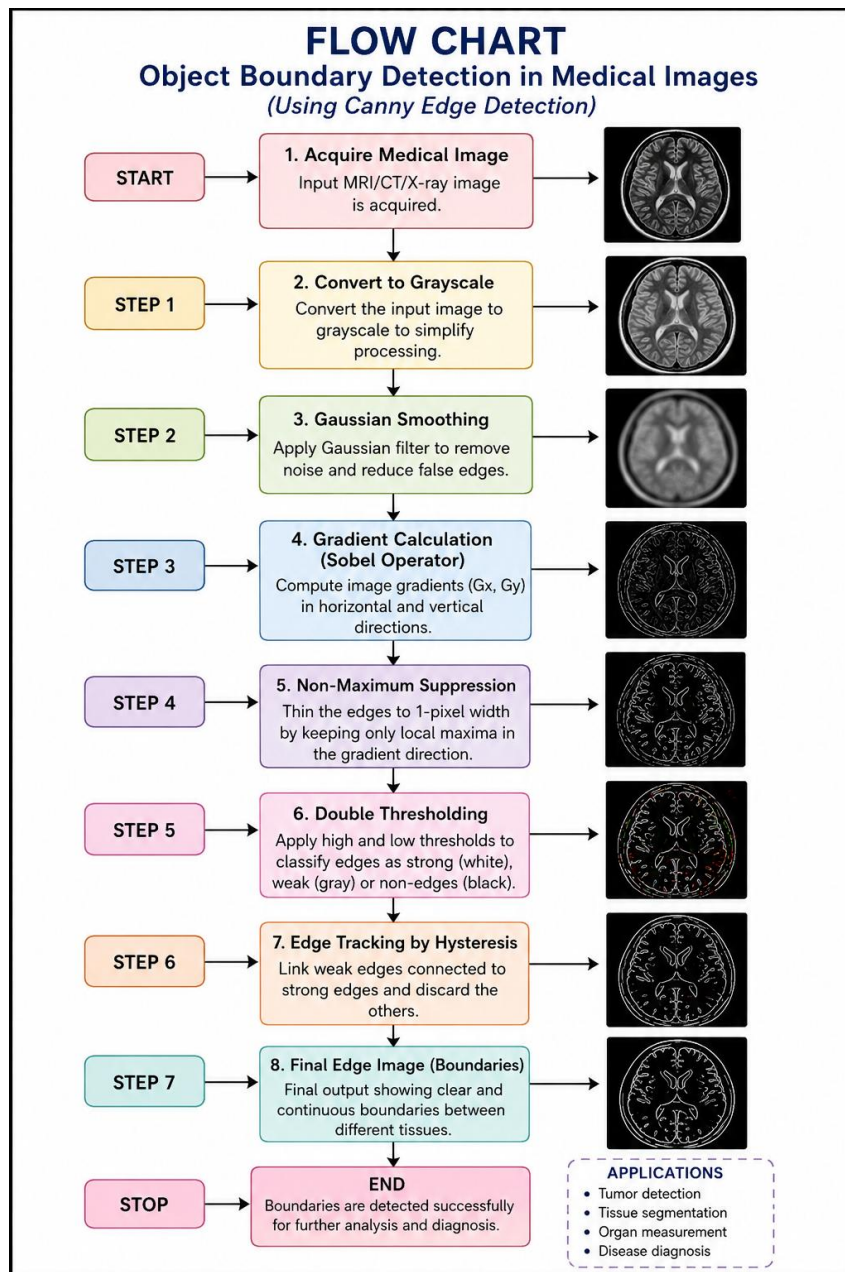


Figure 2: Flow diagram

Expected Result:

- Clear and continuous tissue boundaries.
- Reduced image noise.
- Improved segmentation accuracy.
- Better visualization of organs and abnormalities.
- Supports accurate diagnosis and treatment planning.

Conclusion:

Boundary detection is an essential process in medical image analysis because it accurately separates different tissues and anatomical structures. The Canny Edge Detection algorithm is the most appropriate technique as it combines noise reduction with precise edge detection. It produces thin, continuous, and reliable boundaries, improving image segmentation and enabling accurate medical diagnosis.